

STRIPOLE- Pantoprazole Powder for Solution for Injection 40mg

Composition:
Each vial contains:

Active Ingredient:
Pantoprazole Sodium Sesquihydrate BP Equivalent to Pantoprazole.....40 mg
(Suitably Buffered)

Excipients:
Disodium Edetate BP1 mg
Sodium Hydroxide BPq.s.
Water for Injection BPq.s.

Dosage Form:
Powder for Solution for Injection (lyophilised powder)

Product Description:
A white or almost white lyophilised cake and/or powder.

Description after Reconstitution:
Clear transparent solution, leaving no visible residue as un-dissolved matter.

Pharmacodynamics:
Mechanism of action
Pantoprazole is a substituted benzimidazole which inhibits the secretion of hydrochloric acid in the stomach by specific action on the proton pumps of the parietal cells.
Pantoprazole is converted to its active form in the acidic environment in the parietal cells where it inhibits the H⁺, K⁺-ATPase enzyme, i.e. the final stage in the production of hydrochloric acid in the stomach. The inhibition is dose-dependent and affects both basal and stimulated acid secretion. In most patients, freedom from symptoms is achieved within 2 weeks. As with other proton pump inhibitors and H₂ receptor inhibitors, treatment with pantoprazole reduces acidity in the stomach and thereby increases gastrin in proportion to the reduction in acidity. The increase in gastrin is reversible. Since pantoprazole binds to the enzyme distal to the cell receptor level, it can inhibit hydrochloric acid secretion independently of stimulation by other substances (acetylcholine, histamine, gastrin). The effect is the same whether the product is given orally or intravenously.

Pharmacodynamic effects
The fasting gastrin values increase under pantoprazole. On short-term use, in most cases they do not exceed the upper limit of normal. During long-term treatment, gastrin levels double in most cases. An excessive increase, however, occurs only in isolated cases. As a result, a mild to moderate increase in the number of specific endocrine (ECL) cells in the stomach is observed in a minority of cases during long-term treatment (simple to adenomatoid hyperplasia). However, according to the studies conducted so far, the formation of carcinoid precursors (atypical hyperplasia) or gastric carcinoids as were found in animal experiments (see section 5.3) have not been observed in humans.

An influence of a long term treatment with pantoprazole exceeding one year cannot be completely ruled out on endocrine parameters of the thyroid according to results in animal studies.

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.

Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics:
General Pharmacokinetics
Pharmacokinetics do not vary after single or repeated administration. In the dose range of 10 to 80 mg, the plasma kinetics of pantoprazole are linear after both oral and intravenous administration.

Distribution
Pantoprazole's serum protein binding is about 98%. Volume of distribution is about 0.15 l/kg.

Biotransformation
The substance is almost exclusively metabolized in the liver. The main metabolic pathway is demethylation by CYP2C19 with subsequent sulphate conjugation, other metabolic pathway include oxidation by CYP3A4.

Elimination
Terminal half-life is about 1 hour and clearance is about 0.1 l/h/kg. There were a few cases of subjects with delayed elimination. Because of the specific binding of pantoprazole to the proton pumps of the parietal cell the elimination half-life does not correlate with the much longer duration of action (inhibition of acid secretion).
Renal elimination represents the major route of excretion (about 80%) for the metabolites of pantoprazole, the rest are excreted with the faeces. The main metabolite in both the serum and urine is desmethylpantoprazole which is conjugated with sulphate. The half-life of the main metabolite (about 1.5 hours) is not much longer than that of pantoprazole.

Special populations
Poor metabolisers
Approximately 3 % of the European population lack a functional CYP2C19 enzyme and are called poor metabolisers. In these individuals the metabolism of pantoprazole is probably mainly catalysed by CYP3A4. After a single-dose administration of 40 mg pantoprazole, the mean area under the plasma concentration-time curve was approximately 6 times higher in poor metabolisers than in subjects having a functional CYP2C19 enzyme (extensive metabolisers). Mean peak plasma concentrations were increased by about 60 %. These findings have no implications for the posology of pantoprazole.

Renal impairment
No dose reduction is recommended when pantoprazole is administered to patients with impaired renal function (incl. dialysis patients). As with healthy subjects, pantoprazole's half-life is short. Only very small amounts of pantoprazole are dialyzed. Although the main metabolite has a moderately delayed half-life (2 - 3 h), excretion is still rapid and thus accumulation does not occur.

Hepatic impairment
Although for patients with liver cirrhosis (classes A and B according to Child) the half-life time values increased to between 7 and 9 h and the AUC values increased by a factor of 5 - 7, the maximum serumconcentration only increased slightly by a factor of 1.5 compared with healthy subjects.

Older people
A slight increase in AUC and C_{max} in elderly volunteers compared with younger counterparts is also not clinically relevant.

Pediatric population
Following administration of single intravenous doses of 0.8 or 1.6 mg/kg pantoprazole to children aged 2 – 16 years there was no significant association between pantoprazole clearance and age or weight. AUC and volume of distribution were in accordance with data from adults.

Indication:
Short term use of symptomatic improvement and healing of gastrointestinal diseases which require a reduction in acid secretion:
• Duodenal ulcer
• Gastric ulcer
• Moderate and severe reflux oesophagitis
• Zollinger-Ellison Syndrome and other pathological hypersecretory conditions

Recommended Dose:
The intravenous administration of Stripole i.v. is recommended only if oral application is not appropriate. The recommended dose for gastric and duodenal ulcer and moderate and severe reflux esophagitis is one vial of 40mg Stripole per day. For the long-term management of Zollinger- Ellison-Syndrome and other pathological hypersecretory conditions the recommended daily dose at the beginning of the treatment is 80mg Stripole. Therefore, the dosage can be titrated up or down as needed using measurements of gastric acid secretion to guide. With doses above 80mg daily, the dose should be divided and given twice daily. A temporary increase of the dosage above 160mg pantoprazole is possible but should not be applied longer than required for adequate acid control. In case a rapid acid control is required, a starting dose of 2 x 80mg Stripole is sufficient to manage a decrease of acid output into the target range (<10 mEq/h) within one hour in the majority of patients. Transition from Stripole to the oral formulation of Stripole should be performed as soon as it is clinically justified. The recommended intravenous dosage is one vial (40mg pantoprazole) Stripole per day.

Instructions for use/handling:
A ready-to-use solution is prepared by injecting 10ml of physiological sodium chloride solution into the vial containing the dry substance. This solution may be administered directly or may be administered after mixing with 100ml physiological sodium chloride solution or 5% glucose. Stripole should not be prepared or mixed with solvents other than those stated. After preparation the solution must be used within 12 hours. From a

microbiological point of view the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 12 hours at not more than 25°C. The drug should be administered intravenously over 2-15 minutes.

Special Patient Populations
Impaired Hepatic Function:
A daily dose of 20mg pantoprazole (half a vial of 40mg pantprazole I.V.) should not be exceeded in patients with severe liver impairment.

Impaired Renal Function:
No dose adjustment is necessary in patients with impaired renal function.Type and duration of treatment:
A soon as oral therapy is possible, treatment with Stripole should be discontinued and 40mg pantoprazole p.o. should be administered instead.

Route of Administration:
Intravenous (IV)

Contraindications:
Hypersensitivity to the active substance, substituted benzimidazoles, or to any of the excipients listed.

Warnings and Precautions:
Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

In presence of alarm symptoms
In the presence of any alarm symptom (e. g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment with pantoprazole may alleviate symptoms and delay diagnosis. Further investigation is to be considered if symptoms persist despite adequate treatment.

Hepatic Impairment
In patients with severe liver impairment, the liver enzymes should be monitored during therapy. In the case of a rise of the liver enzymes, the treatment should be discontinued.

Co-administration with atazanavir
Co-administration of atazanavir with proton pump inhibitors is not recommended. If the combination of atazanavir with a proton pump inhibitor is judged unavoidable, close clinical monitoring (e.g. virus load) is recommended in combination with an increase in the dose of atazanavir to 400 mg with 100 mg of ritonavir. A pantoprazole dose of 20 mg per day should not be exceeded.

Gastrointestinal Infections Caused by Bacteria
Pantoprazole, like all proton pump inhibitors (PPIs), might be expected to increase the counts of bacteria normally present in the upper gastrointestinal tract. Treatment with Pantoprazole may lead to a slightly increased risk of gastrointestinal infections caused by bacteria such as Salmonella and Campylobacter or C. difficile.

Hypomagnesaemia
Severe hypomagnesaemia has been reported in patients treated with PPIs like pantoprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPIs with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Subacute Cutaneous Lupus Erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of SCLE. If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping pantoprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Bone fractures
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10–40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Interference with Laboratory Tests
Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. To avoid this interference, pantoprazole treatment should be stopped for at least 5 days before CgA measurements. If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Clostridium Difficile Diarrhea
Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency
Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interactions with Other Medicaments:
Medicinal products with pH Dependent Absorption Pharmacokinetics
Because of profound and long lasting inhibition of gastric acid secretion, pantoprazole may reduce the absorption of drugs with a gastric pH dependent bioavailability, e.g. some azole antifungals as ketoconazole, itraconazole, posaconazole and other medicine as erlotinib.

HIV Protease Inhibitors
Co-administration of atazanavir and other HIV medications whose absorption is pH-dependent with proton-pump inhibitors might result in a substantial reduction in the bioavailability of these HIV medications and might impact the efficacy of these medicines. Therefore, the co-administration of proton pump inhibitors with atazanavir is not recommended.

If the combination of HIV protease inhibitors with a proton pump inhibitor is judged unavoidable, close clinical monitoring (e.g. virus load) is recommended. A pantoprazole dose of 20 mg per day should not be exceeded. Dosage of the HIV protease inhibitor may need to be adjusted.

Coumarin Anticoagulants (phenprocoumon or warfarin)
Although no interaction during concomitant administration of phenprocoumon or warfarin has been observed in clinical pharmacokinetic studies, a few isolated cases of changes in International Normalised Ratio (INR) have been reported during concomitant treatment in the post- marketing period. Increases in INR and prothrombin time may lead to abnormal bleeding, and even death. Therefore, in patients treated with coumarin anticoagulants (e.g. phenprocoumon or warfarin), monitoring of prothrombin time/INR is recommended after initiation, termination or during irregular use of pantoprazole.

Other Interactions Studies
Pantoprazole is extensively metabolised in the liver via the cytochrome P450 enzyme system. The main metabolic pathway is demethylation by CYP2C19 and other metabolic pathways include oxidation by CYP3A4.

Interaction studies with drugs also metabolized with these pathways, like carbamazepine, diazepam, glibenclamide, nifedipine, and an oral contraceptive containing levonorgestrel and ethinyl oestradiol did not reveal clinically significant interactions.

An interaction of pantoprazole with other medicinal products or compounds, which are metabolized using the same enzyme system, cannot be excluded.

There were no interactions with concomitantly administered antacids.

Medicinal Products that Inhibit or Induce CYP2C19
Inhibitors of CYP2C19 such as fluvoxamine could increase the systemic exposure of pantoprazole. A dose reduction may be considered for patients treated long-term with high doses of pantoprazole, or those with hepatic impairment.
Enzyme inducers affecting CYP2C19 and CYP3A4 such as rifampicin and St John's wort (Hypericum perforatum) may reduce the plasma concentrations of PPIs that are

Size:- 150mm x 350mm
Front

	Prepared by:		Reviewed by				Approved BY
Product : Stripole Injection 40mg	PDC	F&D	Regulatory affairs	Production	Quality Control	Quality Assurance	Head –Q-A
Component: Packinsert Dimension: 150 x 350mm (front & back) Colour: Black Version: GSM: 54 gsm Maplitho paper Varnish:							
	Sign/Date	Sign/Date	Sign/Date	Sign/Date	Sign/Date	Sign/Date	Sign/Date
Remark:							

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metabolized through these enzyme systems.

Methotrexate

Concomitant use of high dose methotrexate (e.g. 300 mg) and proton pump inhibitors has been reported to increase methotrexate levels in some patients. Therefore in settings where high-dose methotrexate is used, for example cancer and psoriasis, a temporary withdrawal of pantoprazole may need to be considered.

Pregnancy and Lactation:

Pregnancy:

A moderate amount of data on pregnant women (between 300-1000 pregnancy outcomes) indicate no malformative or feto/ neonatal toxicity of pantoprazole. Studies in animals have shown reproductive toxicity. The potential risk for humans is unknown. Pantoprazole should not be used during pregnancy unless clearly necessary.

Breastfeeding:

Animal studies have shown excretion of pantoprazole in breast milk. There is insufficient information on the excretion into human milk but excretion into human milk has been reported. A risk to the newborn/infant cannot be excluded. Therefore, a decision on whether to continue/discontinue breast-feeding or to continue/discontinue therapy with Pantoprazole should be made taking into account the benefit of breast-feeding to the child and the benefit of Pantoprazole therapy to woman.

Fertility

There was no evidence of impaired fertility.

Side Effects:

Approximately 5% of patients can be expected to experience adverse drug reactions (ADRs). The most commonly reported ADRs are diarrhea and headache, both occurring in approximately 1% of patients.

The table below lists adverse reactions reported with pantoprazole, ranked under the following frequency classification:

Very common;

Common;

Uncommon;

Rare;

Very Rare;

not known (cannot be estimated from the available data).

For all adverse reactions reported from post-marketing experience, it is not possible to apply any Adverse Reaction frequency and therefore they are mentioned with a "not known" frequency.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1. Adverse reactions with pantoprazole in clinical trials and post- marketing experience

Frequency	Common	Uncommon	Rare	Very Rare	Not Known
System Organ Class					
Blood and lymphatic system disorders			Agranulocytosis	Thrombocytopenia; Leukopenia; Pancytopenia	
Immune system disorders			Hypersensitivity (including anaphylactic reactions and anaphylactic shock)		
Metabolism and nutrition disorders			Hyperlipidaemias and lipid increases (triglycerides, cholesterol); Weight changes		Hyponatraemia; Hypomagnesaemia; Hypocalcaemia ¹⁾ ; Hypokalaemia.
Psychiatric disorders		Sleep disorders	Depression (and all aggravations)	Disorientation (and all aggravations)	Hallucination; Confusion (especially in pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence)
Nervous system disorders		Headache, Dizziness	Taste disorders		Parasthesia
Eye disorders			Disturbance in vision/ blurred vision		
Gastrointestinal disorders	Fundic gland polyps (benign)	Diarrhoea; Nausea/ vomiting; Abdominal distension and bloating; Constipation; Dry mouth; Abdominal pain and discomfort			Microscopic colitis
Hepatobiliary disorders		Liver enzymes increased (transaminases, γ-GT)	Bilirubin increased		Hepatocellular injury; Jaundice; Hepatocellular failure
Skin and subcutaneous tissue disorders		Rash/ exanthema/ eruption; Pruritus	Urticaria; Angioedema		Stevens- Johnson syndrome; Lyell syndrome; Erythema multiforme; Photosensitivity; Subacute cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders		Fracture of the hip, wrist or spine	Arthralgia; Myalgia		Muscle spasm ²⁾
Renal and urinary disorders					Interstitial nephritis (with possible progression to renal failure)
Reproductive system and breast disorders			Gynaecomastia		
General disorders and administration site conditions	Injection site thrombophlebitis	Asthenia, fatigue and malaise	Body temperature increased; Oedema peripheral		

1. Hypocalcaemia in association with hypomagnesaemia
2. Muscle spasm as a consequence of electrolyte disturbance

Subacute Cutaneous Lupus Erythematosus (SCLE)

Skin and subcutaneous tissue disorders

Frequency "not known": Subacute cutaneous lupus erythematosus

Interstitial Nephritis

Renal and urinary disorders: Interstitial nephritis

Hypomagnesaemia

Metabolism and nutritional disorders

Frequency "not known": hypomagnesaemia

Fracture

Musculoskeletal disorders

Frequency "uncommon": Fracture of the hip, wrist or spine

Clostridium Difficile Diarrhoea

Infections & infestations: Clostridium difficile associated diarrhoea

Fundic Gland Polyps (Benign)

Gastrointestinal disorders

Frequency "common": Fundic gland polyps (benign)

Vitamin B12 Deficiency

Metabolic/Nutritional: Vitamin B12 deficiency

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

Symptoms and Treatment of Overdose:

There are no known symptoms of overdose in man. Systemic exposures with up to 240 mg administered intravenously over 2 minutes were well tolerated. As pantoprazole is extensively protein bound, it is not readily dialysable. In case of overdose with clinical signs of intoxication, apart from symptomatic and supportive treatment, no specific therapeutic recommendations can be made.

Effect on Ability to Drive and Use Machine:

Pantoprazole has no or negligible influence on the ability to drive and use machines.

Adverse drug reactions such as dizziness and visual disturbances may occur. If affected, patients should not drive or operate machines.

Preclinical Safety Data:

Not available.

Incompatibilities:

This medicinal product must not be mixed with other medicinal products except solvents such as sodium chloride 0.9% solution for injection or glucose 5% solution for injection.

Storage Condition:

Store below 30°C. Protect from light.

Reconstituted Solution:

Store at 25°C after reconstitute.

Shelf Life:

24 months

Reconstituted Solution:

12 hours

ATC Code:

A02BC02 –PANTOPRAZOLE

Presentation:

One or ten vials of 10ml clear glass vial packed in a carton long with leaflet

Registration Number:

MAL20076035AZ

Manufacturer:

Gufic Biosciences Limited, Unit-2
Survey No.: 171, N.H. No. 8, Near
Grid, Kabilpore,
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Product Registration Holder:

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Date of Revision:

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